

Tools of the Trade: Linux and SQL

Module 1
Introduction to operating systems

Get started with the course

Introduction to Course 4
Video • 2 min

Course 4 overview
Reading • 10 min

Helpful resources and tips
Reading • 8 min

The wonderful world of operating systems

Welcome to module 1
Video • 52 sec

Kim: My journey into computing
Video • 1 min

Introduction to operating systems
Video • 2 min

Compare operating systems
Reading • 8 min

Test your knowledge: The wonderful world of operating systems
Practice Assignment • Grade: 100%

The operating system at work

Inside the operating system
Video • 3 min

Requests to the operating system
Reading • 8 min

Practice: Order of operations in the OS
Ungraded Plugin • 10 min

Resource allocation via the OS
Video • 2 min

Virtualization technology
Reading • 8 min

Test your knowledge: The operating system at work
Practice Assignment • Grade: 100%

The user interface

GUI versus CLI
Video • 4 min

The command line in use
Reading • 4 min

Activity: Select the user interface
Practice Assignment • 10 min

Ellen: My path into cybersecurity
Video • 2 min

Test your knowledge: The user interface
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Review: Introduction to operating systems

Wrap-up
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Glossary terms from module 1
Reading • 2 min

Module 1 challenge
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Module 2
The Linux operating system

Module 3
Linux commands in the Bash shell

Module 4
Databases and SQL

The command line in use

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Previously, you explored graphical user interfaces (GUI) and command-line interfaces (CLI). In this reading, you'll compare these two interfaces and learn more about how they're used in cybersecurity.

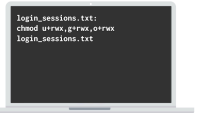
CLI vs. GUI

A **graphical user interface (GUI)** is a user interface that uses icons on the screen to manage different tasks on the computer. A **command-line interface (CLI)** is a text-based user interface that uses commands to interact with the computer.

Display

One notable difference between these two interfaces is how they appear on the screen. A GUI has graphics and icons, such as the icons on your desktop or taskbar for launching programs. In contrast, a CLI only has text. It looks similar to lines of code.


Graphical user interface (GUI)


Command-line interface (CLI)

Function

These two interfaces also differ in how they function. A GUI is an interface that only allows you to make one request at a time. However, a CLI allows you to make multiple requests at a time.

Advantages of a CLI in cybersecurity

The choice between using a GUI or CLI is partly based on personal preference, but security analysts should be able to use both interfaces. Using a CLI can provide certain advantages.

Efficiency

Some prefer the CLI because it can be used more quickly when you know how to manage this interface. For a new user, a GUI might be more efficient because they're easier for beginners to navigate.

Because a CLI can accept multiple requests at one time, it's more powerful when you need to perform multiple tasks efficiently. For example, if you had to create multiple new files in your system, you could quickly perform this task in a CLI. If you were using a GUI, this could take much longer, because you have to repeat the same steps for each new file.

History file

For security analysts, using the Linux CLI is helpful because it records a history file of all the commands and actions in the CLI. If you were using a GUI, your actions are not necessarily saved in a history file.

For example, you might be in a situation where you're responding to an incident using a playbook. The playbook's instructions require you to run a series of different commands. If you used a CLI, you'd be able to go back to the history and ensure all of the commands were correctly used. This could be helpful if there were issues using the playbook and you had to review the steps you performed in the command line.

Additionally, if you suspect an attacker has compromised your system, you might be able to trace their actions using the history file.

Key takeaways

GUIs and CLIs are two types of user interfaces that security analysts should be familiar with. There are multiple differences between a GUI and a CLI, including their displays and how they function. When working in cybersecurity, a CLI is often preferred over a GUI because it can handle multiple tasks simultaneously and it includes a history file.

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This reading explores the differences between Graphical User Interfaces (GUIs) and Command-Line Interfaces (CLIs) and their applications in cybersecurity.

GUI vs. CLI Fundamentals

- A GUI uses icons and graphics for managing computer tasks, familiar from desktop environments.
- A CLI is a text-based interface that relies on commands to interact with the computer.

Functional and Display Differences

- GUIs are visually rich with icons and graphics, while CLIs are text-only, resembling lines of code.
- GUIs typically handle one request at a time, whereas CLIs can process multiple requests simultaneously.

Advantages of CLI in Cybersecurity

- CLIs offer efficiency for experienced users, especially when performing multiple tasks, such as creating numerous files.
- The Linux CLI maintains a history file of all commands, which is crucial for security analysts to review actions during incident response or to trace potential attacker activity.

And if you want to continue exploring this topic, try one of these follow-up questions:

What is the main difference between GUI and CLI displays?

How can you use CLI to perform multiple tasks efficiently?

Why is the CLI history file important in cybersecurity incident response?

Ask me anything

Go to next item

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<https://www.coursera.org/learn/linux-and-sql/supplement/7iLDe/the-command-line-in-use>

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